

A Sensitivity-Gated Cascade for Compute-Efficient Spinal Lesion Detection in Resource-Constrained Settings

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Motivation

Spinal radiographs serve as the primary screening modality for a wide spectrum of spinal pathologies [1-2]. This visual ambiguity induces high inter-observer variability and elevates the risk of delayed diagnoses, which can precipitate irreversible neurological sequelae.

Diagnostic Challenges

leading to missed diagnoses and delayed patient care.

Resource-Constrained Triage

runs fast screening on standard clinic CPUs, filtering normal X-rays to reserve scarce MRI slots for critical patients.

Gaps in Current Approaches

1 Sensitivity-Specificity Trade-Off

Monolithic models trigger high false positives or miss critical subtle lesions under data skew [3]

2 Extreme Class Imbalance

Minute pathologies (<1% FOV) are easily obscured by bone superposition and dominant classes [4-5]

3 High Computational Overhead

Running heavy detectors on normal screening X-rays prevents CPU deployment in low-resource clinics [1,6]

Contributions

Sensitivity-Gated Cascade DERNet Framework

Heterogeneous ensemble of DenseNet121, EfficientNetV2-S, and ResNet50 with weighted soft voting, achieving 91.03% AUROC to filter normal studies.

Attention-Guided Localization

Customized YOLO11-L with CSP-Darknet and C2PSA attention to overcome severe class imbalance and localize minute lesions, achieving 40.10% mAP@0.5.

Transparent & Compute-Efficient Inference

Integrates multi-modal explainability using saliency maps, Grad-CAM, and LIME, with quantitative validation against authentic lesion regions, coupled with a staged pipeline optimized for practical CPU deployment in resource-constrained settings.

Dataset



Fig. 1. Illustration of the heterogeneity and visual subtlety of seven lesions.

We utilized **VinDr-SpineXR** benchmark dataset, comprising 10,466 spinal radiographs partitioned into 8,389 training and 2,077 independent testing images across seven expert-annotated lesion categories.

Methodology

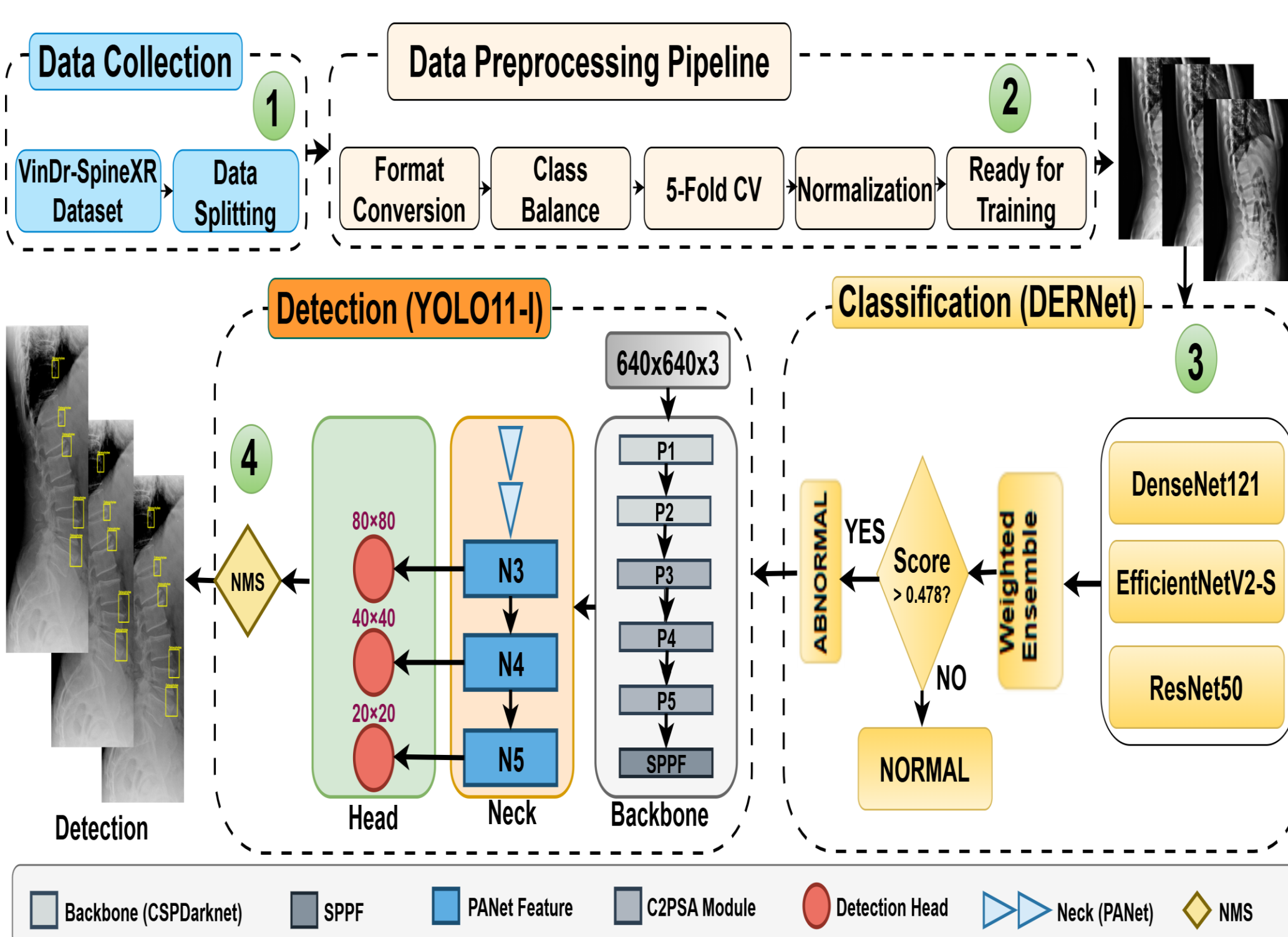


Fig. 2. Architecture of the cascaded framework for spinal lesion triage and localization.

We evaluated our proposed framework on the VinDr-SpineXR benchmark. To resolve inter-scanner exposure disparities and enhance minute lesion contrast, input studies are adaptively preprocessed using CLAHE.

To mitigate radiographic exposure variance CLAHE computes the enhanced intensity by mapping the local CDF as

$$I'(i, j) = \beta \left(\frac{CDF_{\Omega_k}(I(i, j)) - CDF_{min}}{|\Omega_k| - CDF_{min}} \right)$$

Tile Grid (Ω_k): 8 x 8 local grid region scaled by $\beta = 255$.

Clip Limit (τ): $\tau = 2.0$ suppresses contrast over-amplification and noise in homogeneous bones before normalization.

DERNet Framework

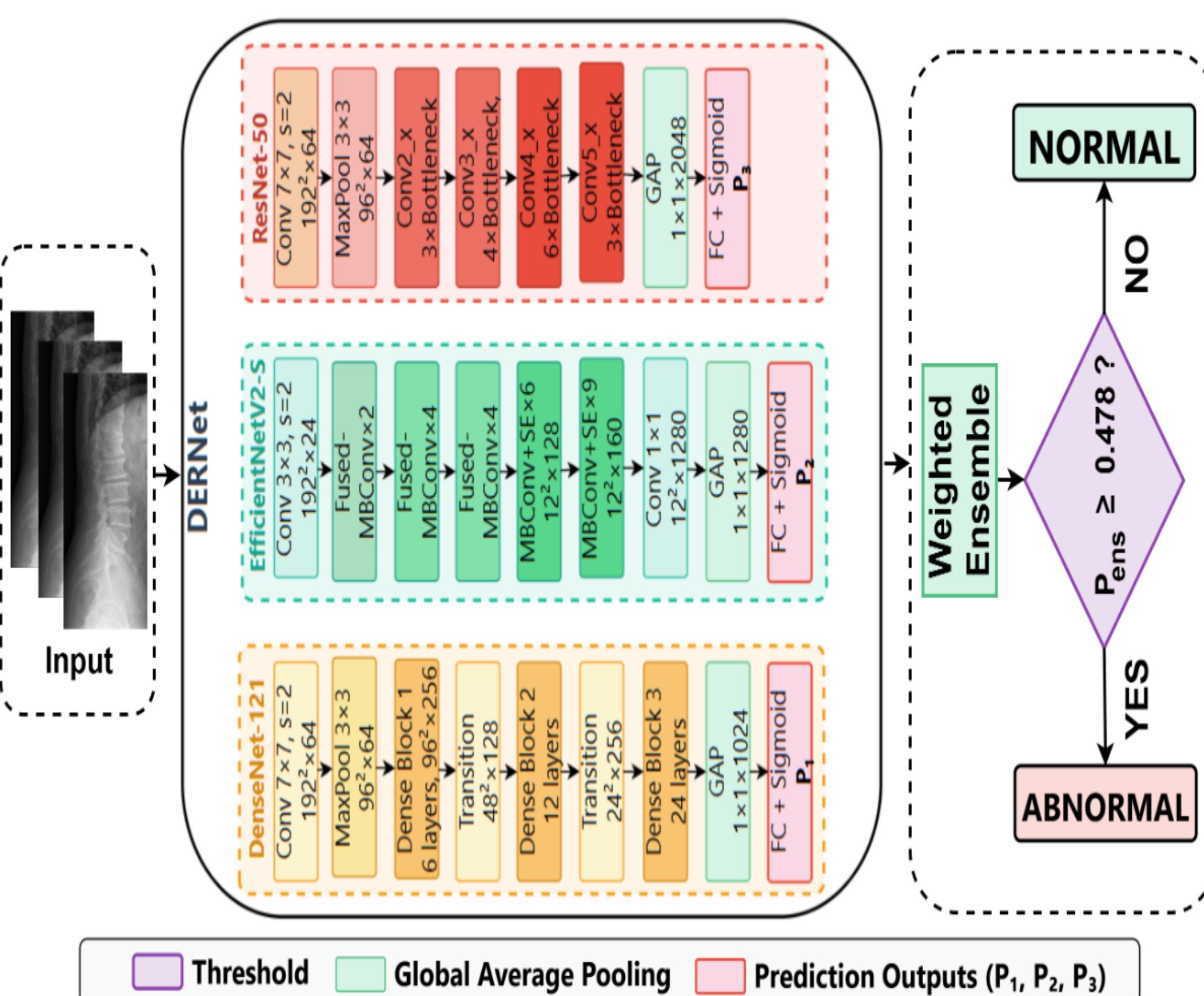


Fig. 3. DERNet probabilistic ensemble architecture for high-sensitivity radiographic triage.

The optimal threshold was found via a grid search with a step size of 0.0002, maximizing the 5-fold cross-validation F1-score on the training data, as

$$\tau^* = \arg \max_{\tau \in [0.35, 0.60]} F_1(\tau)$$

$$F_1(\tau) = \frac{2TP(\tau)}{2TP(\tau) + FP(\tau) + FN(\tau)}$$

Optimal Threshold (τ^*): Grid search across [0.35, 0.60] yields $\tau^* = 0.4787$ to maximize 5-fold CV.

Triage Decision Rule: Studies with $P_{ens} \geq 0.4787$ are routed to YOLO11-L as abnormal; others are filtered as normal

Results

Table 1. Classification performance comparison

Method	AUROC	F1 (%)	Sens. (%)	Spec. (%)
EfficientNetV2-S	89.44	79.34	70.80	91.12
ResNet50	88.88	80.15	82.72	78.13
DenseNet121	86.93	79.55	80.39	79.32
VinDr Ensemble	88.61	81.06	83.07	79.32
HealNNET	88.84	81.20	-	-
DERNet	91.03	83.09	84.91	81.68

DERNet framework outperforms established baselines, achieving a state-of-the-art AUROC of 91.03%.

Table 2. The detection performance for each type of spinal lesion.

Method	LT2 (*)	LT4	LT6	LT8	LT10	LT11	LT13	mAP @0.5
RetinaNet	14.53	25.35	41.67	32.14	65.49	51.85	5.30	28.09
Faster R-CNN	22.66	35.99	49.24	31.68	65.22	51.68	2.16	31.83
Sparse R-CNN	20.09	32.67	48.16	45.32	72.20	49.30	5.41	33.15
VinDr-SpineXR	21.43	27.36	34.78	41.29	62.53	43.39	4.16	33.56
EGCA-Net	22.36	29.75	36.73	44.69	66.58	50.41	2.09	36.09
YOLO12	24.60	38.90	38.10	40.20	68.40	55.50	3.04	38.40
YOLO26	23.60	34.90	34.40	37.50	67.30	43.30	3.35	35.50
Ours (YOLO11-L)	26.70	41.40	40.60	54.80	74.10	51.20	2.99	40.10

(*) LT2, LT4, LT6, LT8, LT10, LT11, and LT13 denote disc space narrowing, foraminal stenosis, osteophytes, spondylolisthesis, surgical implant, vertebral collapse, and other lesions, respectively.

Our customized YOLO11-L outperforms existing detection baselines on VinDr-SpineXR.

AI Explainability (XAI)

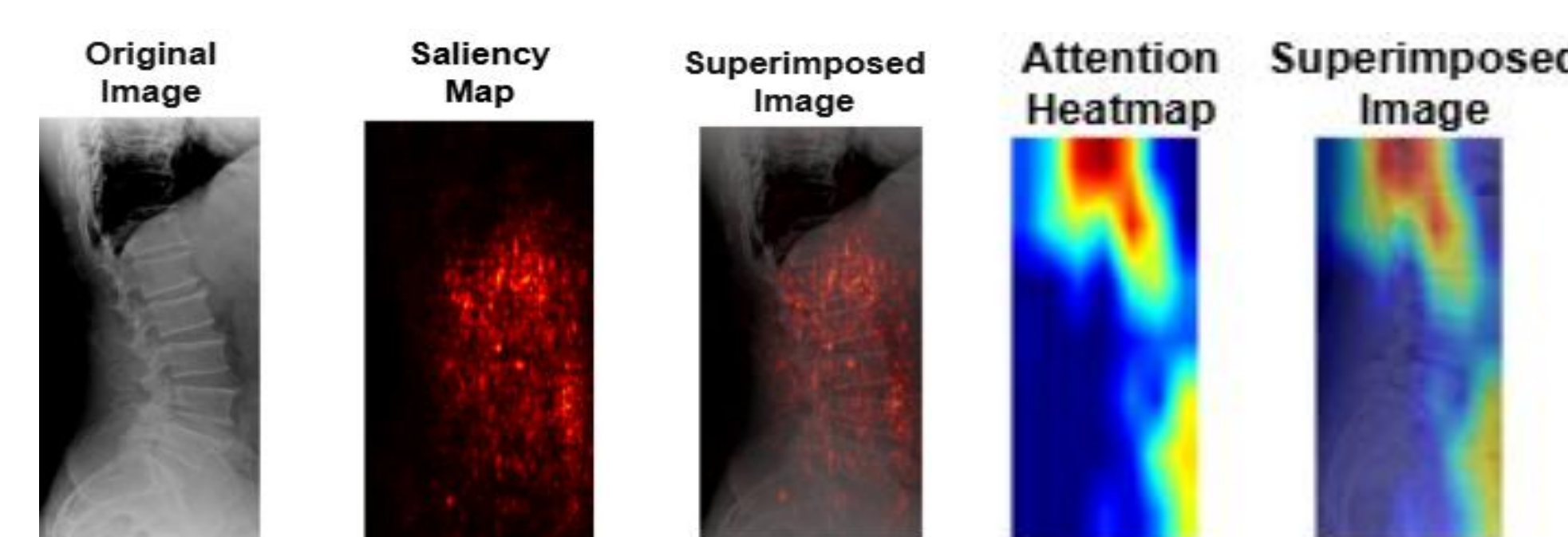


Fig. 4. Multi-modal XAI validation demonstrates attention on genuine lesion pathology.

XAI confirms anatomical grounding, with X-IoU values of 0.61–0.64, indicating that predictions are anchored to authentic spinal pathology rather than artifacts.

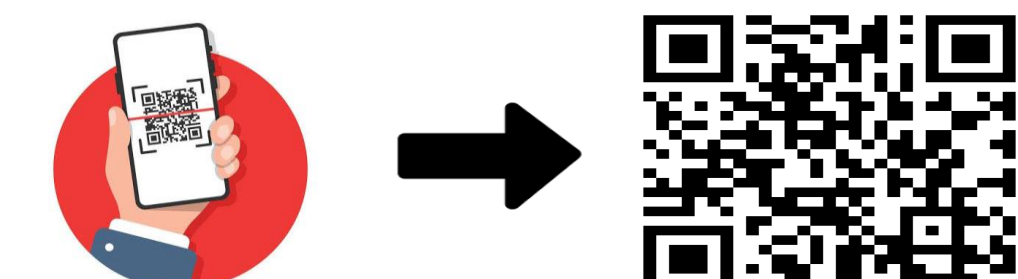
Efficiency and Ablation

CPU triage	15.4 GFLOPS screening filters normals upfront; heavier 74.6 GFLOPS detector executes on abnormalities for CPU triage.
Imbalance ablation	20% copy-paste lifts mAP@0.5 from 37.70% → 40.10% on rare lesions.
Anatomical trust	0.61–0.64 mean X-IoU grounds triage to bone margins, safeguarding scarce MRI slots.

Conclusion

Decouples screening triage from localization via a sensitivity-gated DERNet and YOLO11-L cascade. Delivers 91.03% AUROC and 40.10% mAP@0.5, quantitatively grounded to authentic bone pathology. Enables CPU-efficient triage in resource-constrained clinics, potentially supporting prioritization of scarce MRI resources.

Scan to learn more about DERNet



References

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